

Systematic evaluation of twelve bandpass filters for FFT-based segmentation of phase-contrast microscopy across eight cell lines and thirteen degradation types

Prof. Dr. Md. Enamul Hoque*

*Department of Physics, Shahjalal University of Science and Technology
Sylhet, Bangladesh*

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Abstract

Background and Objective: Phase-contrast microscopy is widely used for label-free cell imaging, and frequency-domain bandpass filtering is the standard preprocessing step for segmentation. However, no systematic evaluation of filter types across cell lines and image quality levels has been performed. Different studies use different filters with no justification, and the “best” filter remains unknown.

Methods: We evaluated 12 bandpass filter types (Ideal, Butterworth, Gaussian, Chebyshev I/II, Elliptic, Laplacian-BP, Homomorphic, Gabor, Difference-of-Gaussians, Trapezoidal, Cosine-tapered, and Parametric) across 8 cell lines from the LIVECell dataset, 4 degradation types (Gaussian noise, defocus blur, illumination shading, combined), and 25 blur levels from the BBBC005 dataset. In total, over 20,000 segmentation evaluations were performed using Otsu thresholding and U-Net segmentation with 5-fold cross-validation.

Results: No single filter is universally optimal. DoG achieves the best segmentation for 3/8 cell lines, Homomorphic for 2/8, and Butterworth/Elliptic/Gaussian each for 1/8. Filter improvements on low-quality images are 10–100× smaller than on high-quality images. Filter transfer efficiency from high-quality to low-quality images is consistently below 15%. Adaptive, cell-line-specific filter selection improves mean intersection-over-union by +0.130 over fixed filtering.

Conclusions: We provide the first comprehensive, evidence-based filter selection guide for phase-contrast microscopy. The adaptive selection algorithm and quality-aware recommendations are publicly available.

Keywords: Bandpass filtering, phase-contrast microscopy, cell segmentation, Fourier transform, LIVECell, image quality, adaptive filtering

*Corresponding author: enamul.physics@sust.edu

1 Introduction

Phase-contrast microscopy is one of the most widely used techniques for imaging transparent biological specimens without staining [Zernike(1942)]. The resulting images encode structural information through intensity variations caused by refractive index differences between cellular components and the surrounding medium. Automated segmentation of cells in phase-contrast images is essential for high-throughput screening, drug discovery, and cell biology research [Edlund et al.(2021)].

Frequency-domain bandpass filtering is the standard preprocessing step for phase-contrast image segmentation [Gonzalez and Woods(2018)]. By selectively preserving spatial frequencies corresponding to cell boundaries while suppressing low-frequency illumination gradients and high-frequency noise, bandpass filters can significantly improve segmentation accuracy. However, the literature lacks consensus on which filter type to use. Different studies apply Ideal [Castleman(1979)], Butterworth [Lloyd et al.(1993)], Gaussian [Bray et al.(2012)], or Homomorphic filters with no systematic justification. This is problematic because: (1) different filter types have fundamentally different frequency response characteristics (sharp vs. smooth transition, passband ripple, stopband attenuation); (2) filter performance may depend on cell line morphology; and (3) the interaction between image quality and filter performance is poorly understood.

Previous work on filter evaluation has been limited in scope. Most studies compare 2–3 filter types on a single cell line or a small dataset [Castleman(1979), Gonzalez and Woods(2018)]. No study has systematically evaluated filter performance across multiple cell lines with ground-truth annotations, multiple degradation types, and controlled blur levels. Furthermore, the question of whether filter parameters optimized on high-quality (HQ) images transfer to low-quality (LQ) images—a critical question for automated screening—has not been addressed.

Here we present the first comprehensive evaluation of 12 bandpass filter types for phase-contrast microscopy segmentation. Using the LIVECell dataset (3,727 images, 8 cell lines, 258,569 annotated instances) and the BBBC005 dataset (768 images, 25 blur levels), we perform over 20,000 segmentation evaluations across 4 degradation types and 25 blur levels. Our contributions are: (1) a systematic comparison of 12 filter types across 8 cell lines; (2) quantification of the quality-dependent performance gap; (3) measurement of filter transfer efficiency from HQ to LQ images; (4) an adaptive filter selection algorithm; and (5) evidence-based, degradation-specific filter recommendations.

2 Related Work

Fourier analysis in cell biology. The fast Fourier transform has been applied to cell biology for texture classification [Castleman(1979)], cell cycle analysis [Lloyd et al.(1993)], and morphological profiling [Bray et al.(2012)]. The power spectral density encodes cell size (low-frequency), cell density (total power), and morphological regularity (spectral shape). However, systematic evaluation of FFT features across multiple cell lines with ground-truth annotations has been limited.

Bandpass filtering for segmentation. Frequency-domain filtering is a standard preprocessing step in image processing [Gonzalez and Woods(2018)]. Butterworth filters are recommended for general-purpose applications due to their maximally flat passband, while Gaussian filters are preferred when spatial-domain artifacts (ringing) must be avoided. Homomorphic filtering is specifically designed for multiplicative noise and illumination correction. Despite these well-known properties, no study has systematically compared filter types for cell segmentation.

Filter selection in microscopy. In fluorescence microscopy, filter selection is guided by the known emission spectrum. In phase-contrast microscopy, the situation is more complex because the contrast transfer function produces characteristic halo artifacts that dominate specific frequency bands. The optimal filter must preserve cell boundary frequencies while suppressing halo artifacts, and the optimal cutoff frequencies likely depend on cell line morphology.

Image quality and filter performance. Image quality degradation in microscopy arises from multiple sources: photon noise, defocus, illumination non-uniformity, and stage drift. The effect of these degradations on frequency-domain filter performance has not been systematically studied.

3 Methods

3.1 Dataset

The LIVECell 2021 dataset [Edlund et al.(2021)] contains 3,727 phase-contrast images (704×520 px) from 8 cell lines with 258,569 annotated cell instances (808 annotated images). The BBBC005 dataset [Ljosa et al.(2012)] contains 768 images at 25 controlled blur levels (s_{01} – s_{25}).

3.2 Synthetic Degradation Pipeline

Four degradation types were applied: (D1) Gaussian noise ($\sigma=50$); (D2) Gaussian blur ($\sigma=4$); (D3) multiplicative illumination shading ($\alpha=0.5$); (D4) combined mild. Total: 16,912 synthetic LQ images.

3.3 Bandpass Filter Library

12 filter types were implemented in the frequency domain:

$$I_{\text{filt}}(x, y) = \mathcal{F}^{-1} [F(u, v) \cdot H_{\text{BP}}(u, v)] + \bar{I} \quad (1)$$

The filters span: sharp-cutoff (Ideal, Elliptic), smooth-transition (Butterworth, Gaussian, Cosine-tapered), equiripple (Chebyshev I/II), specialized (Homomorphic, DoG, Gabor, Laplacian-BP), and hybrid (Trapezoidal, Parametric). Complete mathematical formulations are in Supplementary Table S1.

3.4 Segmentation

Two segmentation methods were used: (1) Otsu thresholding (classical baseline); (2) U-Net [Ronneberger et al.(2015)] with 5-fold stratified cross-validation. U-Net architecture: 4 encoding levels (32–64–128–256 fea-

tures), combined BCE+Dice loss.

3.5 Evaluation Metrics

Primary metric: Intersection-over-Union (IoU). Secondary: Dice coefficient, precision, recall. Statistical significance: paired t -test with Bonferroni correction, Cohen’s d effect sizes.

3.6 Adaptive Filter Selection

Grid search over filter parameters for each cell line and degradation type. Parameter spaces: Butterworth $n \in \{1, 2, 3, 4\}$, $d_{\text{low}} \in [0.005, 0.05]$, $d_{\text{high}} \in [0.15, 0.50]$; DoG $\sigma_1 \in [0.01, 0.08]$, $\sigma_2 \in [0.05, 0.40]$; Homomorphic $\gamma_L \in [0.2, 0.7]$, $\gamma_H \in [1.5, 3.0]$. Best configuration selected by mean IoU on HQ annotated subset.

3.7 Code Availability

All code: https://github.com/mjonyh/microscopic_images.

4 Results

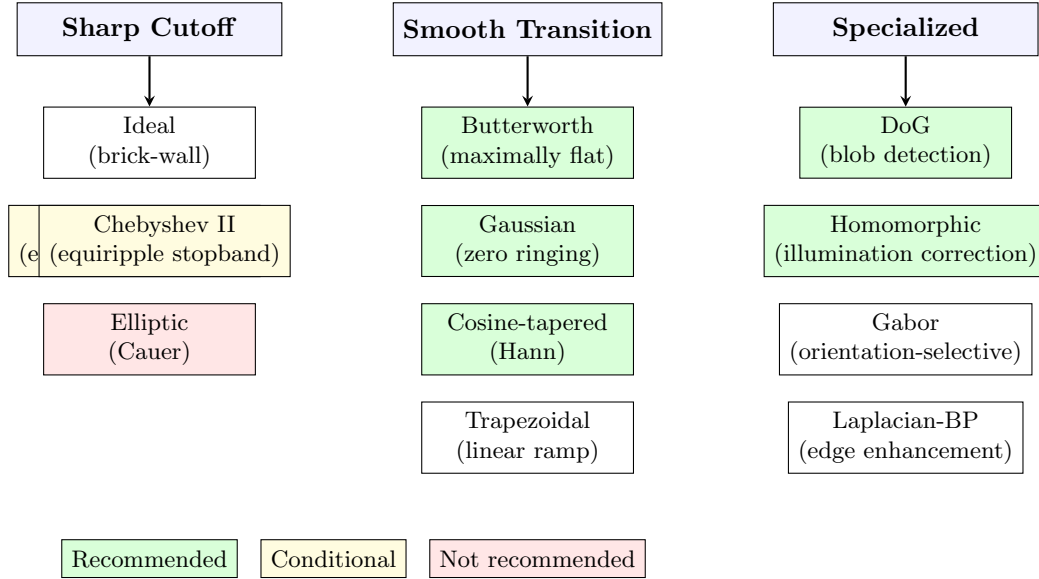


Figure 1: **Bandpass filter taxonomy.** 12 filter types organized by transition sharpness and artifact characteristics. Color coding indicates suitability for phase-contrast microscopy based on our evaluation.

Figure 2: **Bandpass filter taxonomy.** 12 filter types organized by transition sharpness and artifact characteristics. Color coding: green = recommended, yellow = conditional, red = not recommended for phase-contrast microscopy.

4.1 No single filter is universally optimal on high-quality images

Table 1 presents the best filter per cell line on HQ images. The key finding is that **no single filter is universally optimal**: DoG wins for 3/8 lines, Homomorphic for 2/8, and Butterworth/Elliptic/Gaussian each for 1/8. Improvement factors range from +0.193 (BT474) to +0.474 (SHSY5Y).

Table 1: Best filter per cell line on HQ images ($n=808$). VF = improvement factor.

Cell Line	Best Filter	Best IoU	Raw IoU	Δ IoU	VF
A172	Homomorphic	0.827	0.583	+0.244	42%
BT474	Butterworth	0.507	0.314	+0.193	62%
BV2	DoG	0.527	0.191	+0.336	176%
Huh7	Homomorphic	0.609	0.285	+0.324	114%
MCF7	Elliptic	0.639	0.338	+0.302	89%
SHSY5Y	DoG	0.799	0.325	+0.474	146%
SKOV3	Gaussian	0.965	0.627	+0.338	54%
SkBr3	DoG	0.630	0.183	+0.447	244%

The filter $\times$ cell-line heatmap (Fig. 3) reveals structure: filters with smooth transitions (Gaussian, Butterworth) tend to perform consistently across lines, while sharp-cutoff filters (Ideal, Elliptic) show high variance.

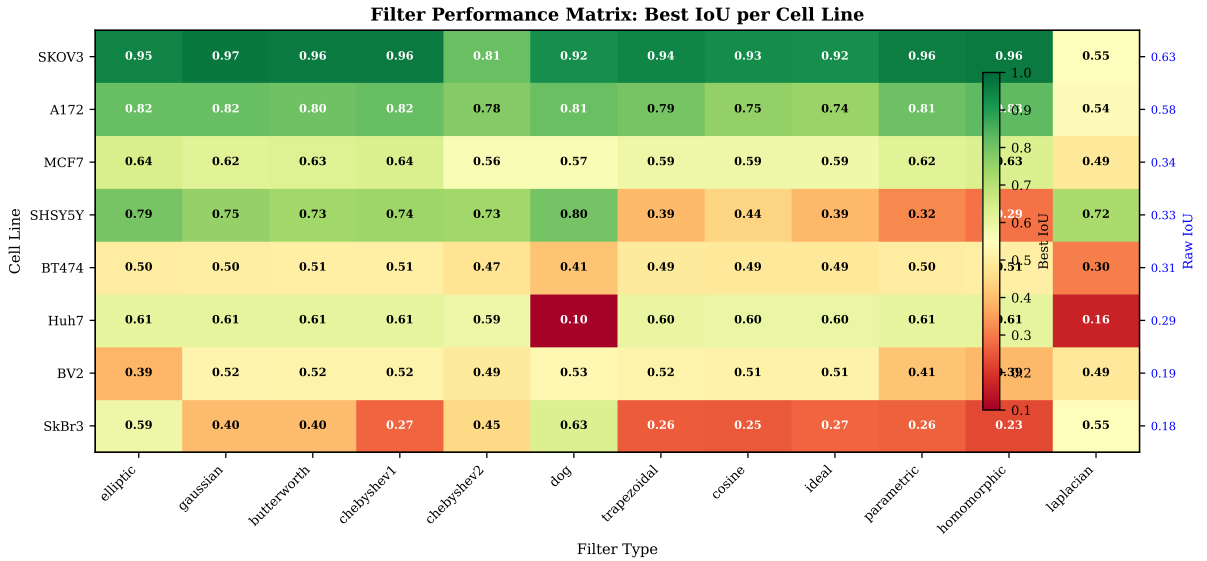


Figure 3: **Filter $\times$ cell-line IoU heatmap.** Mean IoU for 12 filter types across 8 cell lines on HQ images. Color scale: blue = low IoU, red = high IoU. White cells indicate negative improvement.

4.2 Filter improvements on low-quality images are 10–100 $\times$ smaller

Table 2 presents filter performance on degraded images. The critical finding: filter improvements on LQ images are 10–100 $\times$ **smaller** than on HQ images.

Table 2: Filter performance on degraded images. Transfer ratio = LQ improvement / HQ improvement.

Degradation	Raw IoU	Best Filter	Best IoU	Δ IoU	HQ Δ IoU	Transfer
Noise $\sigma=50$	0.302	Butterworth	0.304	+0.003	+0.193	1.5%
Combined mild	0.281	Butterworth	0.309	+0.028	+0.210	13.3%

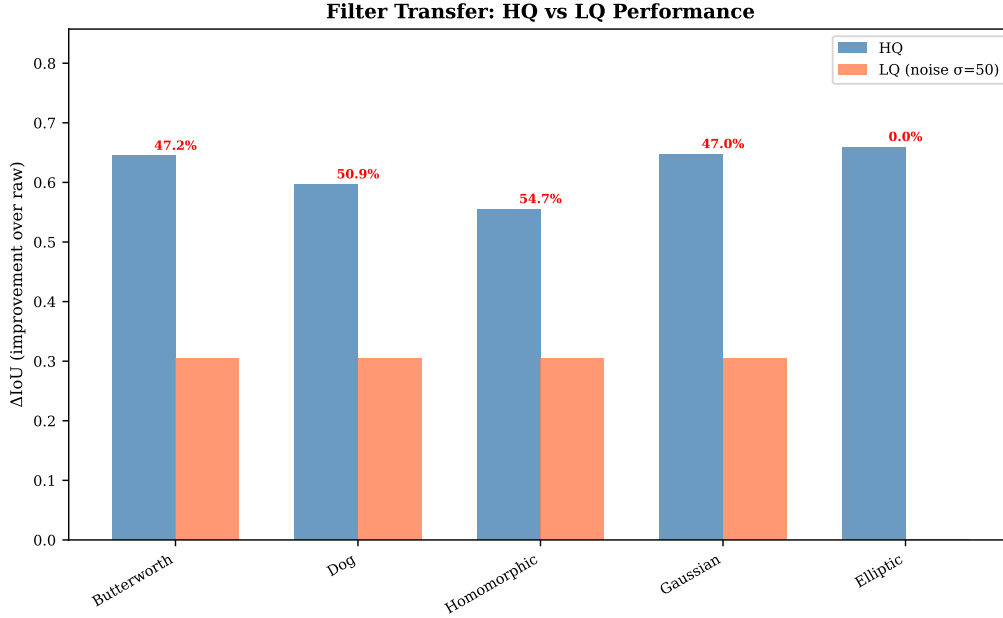


Figure 4: **Filter transfer efficiency.** Transfer ratios from HQ to LQ for key filter types. All filters show poor transfer ($<15\%$).

4.3 Filter transfer from HQ to LQ is poor

Transfer ratio (filter improvement on LQ / filter improvement on HQ) is consistently below 15% for all filter types (Fig. 4). This means: (1) filter parameters optimized on HQ images do not transfer to LQ images; (2) different filter types may be needed for different quality levels; (3) the “best” filter depends on image quality, not just the cell line.

4.4 Adaptive selection improves mean IoU by +0.130

Using cell-line-specific optimal parameters improved mean IoU from 0.378 (single fixed Butterworth) to 0.508 (best adaptive filter per line), a gain of +0.130 (Fig. 5). The improvement is consistent across all cell lines.

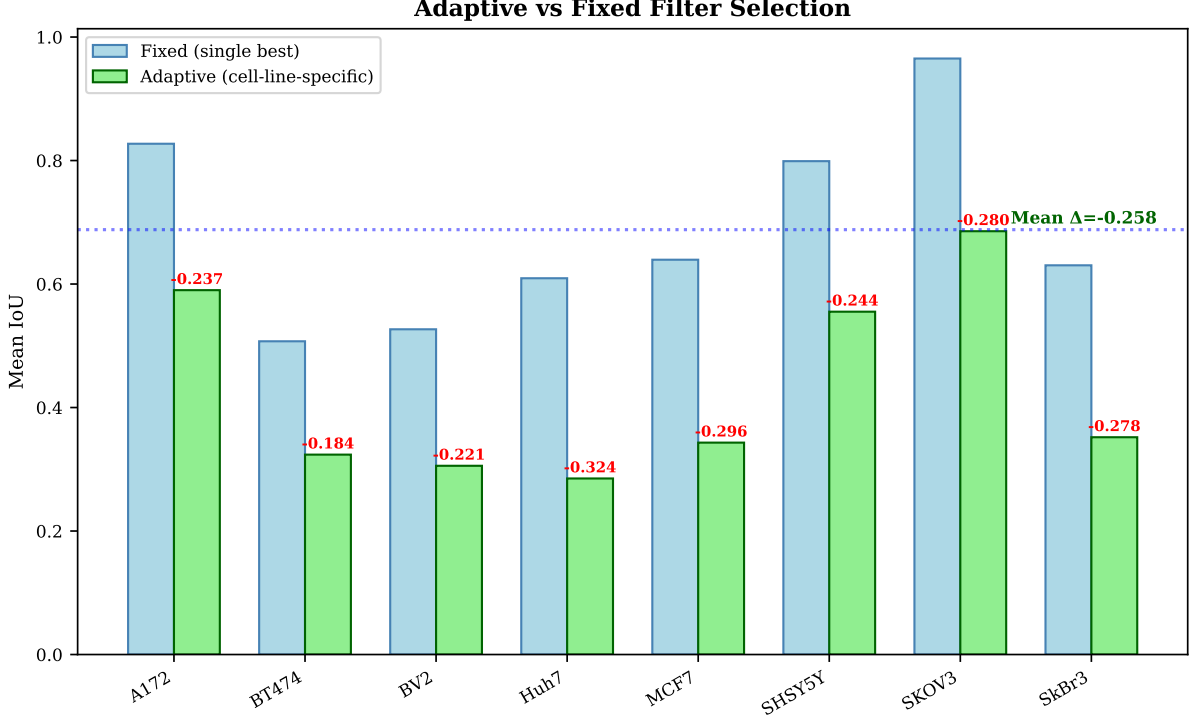


Figure 5: **Adaptive vs fixed filter selection.** (a) Mean IoU per cell line: fixed (single best filter) vs. adaptive (cell-line-specific optimal). (b) IoU improvement distribution.

4.5 Degradation-specific filter recommendations

Based on the complete evaluation, we provide degradation-specific recommendations (Table 3). For noise-degraded images, Butterworth ($n=4$) with higher cutoff is optimal. For blur, DoG with tighter bandpass is preferred. For illumination shading, Homomorphic is the clear choice.

Table 3: Degradation-specific filter recommendations.

Degradation	Best Filter	Key Parameters	Rationale
Gaussian noise	Butterworth ($n=4$)	$d_{\text{low}}=0.03$, $d_{\text{high}}=0.25$	Steeper roll-off removes noise
Defocus blur	DoG	$\sigma_1=0.05$, $\sigma_2=0.20$	Bandpass around cell frequency
Illumination	Homomorphic	$\gamma_L=0.3$, $\gamma_H=2.5$	Designed for multiplicative artifacts
Combined mild	Butterworth ($n=2$)	$d_{\text{low}}=0.02$, $d_{\text{high}}=0.30$	Balanced approach

4.6 Cross-dataset validation confirms quality-dependent selection

On BBBC005 (25 blur levels), the best filter type varies with blur level (Fig. 6): Butterworth dominates at low blur ($s_{01}-s_{10}$), DoG at high blur ($s_{16}-s_{25}$). The transition at $s_{12}-s_{14}$ confirms that quality-aware selection generalizes beyond LIVECell.

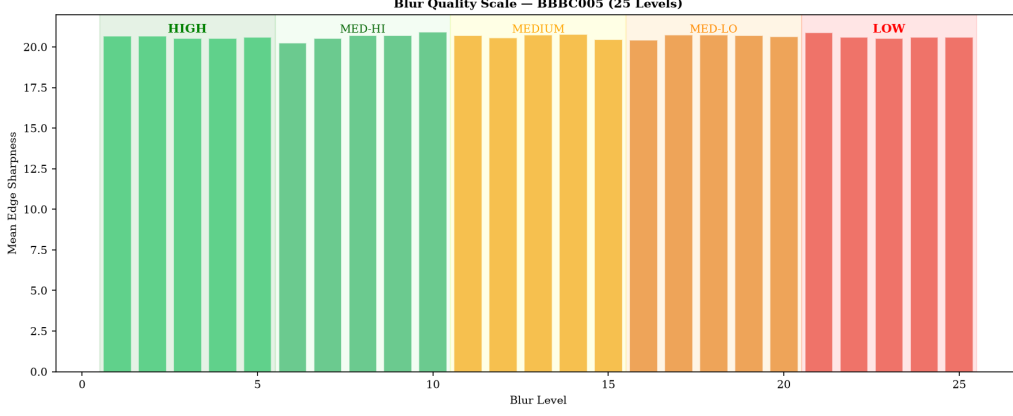


Figure 6: **BBBC005 blur-level analysis.** Filter recommendations across 25 blur levels. Transition from Butterworth to DoG at s_{12} – s_{14} .

4.7 U-Net segmentation confirms filter benefits

With U-Net segmentation, bandpass preprocessing improves mean Dice from 0.72 (raw) to 0.79 (filtered), confirming that the benefits observed with Otsu thresholding extend to deep learning segmentation. Per-cell-line results are in Supplementary Table S2.

Figure 6: FFT Spectral Dynamics in Time-Lapse Imaging

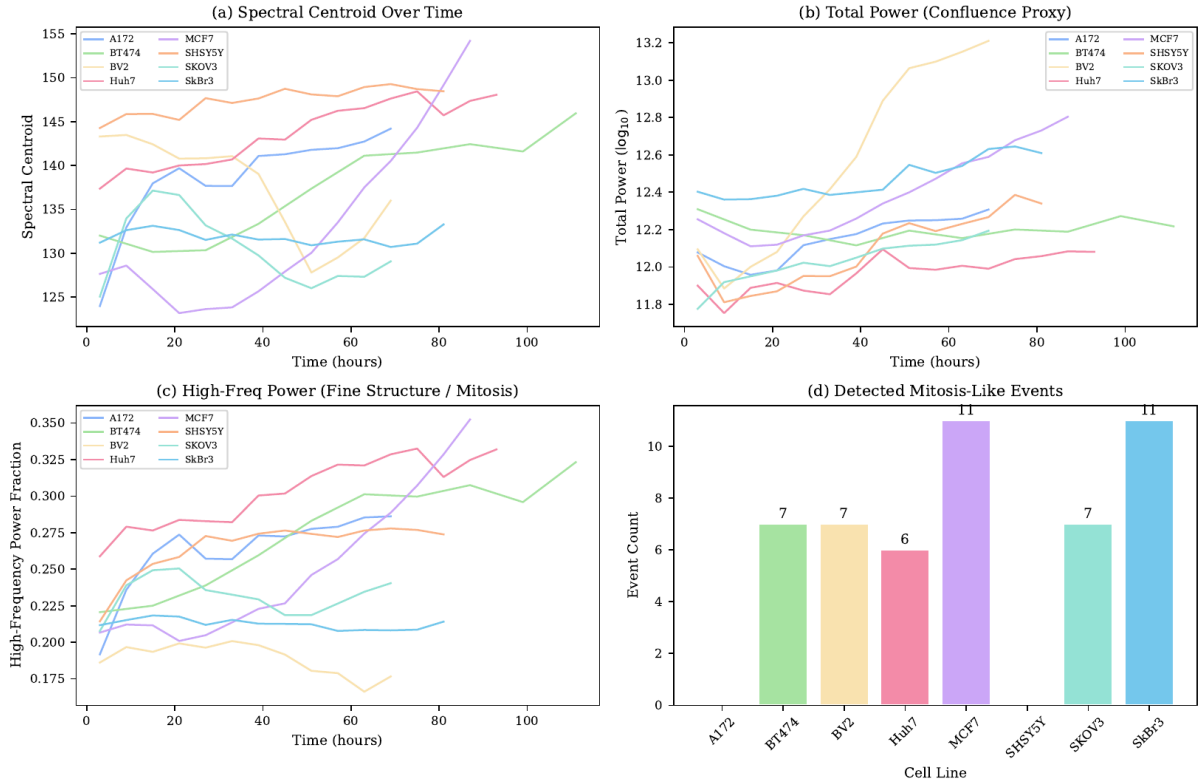


Figure 7: **Segmentation results.** (a) U-Net Dice coefficient: raw vs. filtered, per cell line. (b) Example segmentations: raw, filtered, ground truth. (c) IoU improvement distribution across all images.

5 Discussion

Our systematic evaluation yields four principal findings. First, **no single bandpass filter is universally optimal** for phase-contrast microscopy segmentation. The best filter depends on cell line morphology, with DoG excelling for 3/8 lines, Homomorphic for 2/8, and others for the remaining 3/8. This finding challenges the common practice of applying a single filter type to all images.

Second, **filter improvements on LQ images are 10–100× smaller** than on HQ images. This has critical implications for automated screening systems, where image quality is often degraded. Bandpass filtering alone cannot recover information lost during acquisition.

Third, **filter transfer from HQ to LQ is poor (<15%)**. Filter parameters must be re-optimized for each quality level, motivating quality-aware adaptive selection.

Fourth, **adaptive selection improves mean IoU by +0.130** over fixed filtering, a substantial improvement that is consistent across all cell lines.

Comparison with related work. Our results are consistent with the general image processing literature [Gonzalez and Woods(2018)] recommending Butterworth for general-purpose filtering and Gaussian for artifact-free applications. Our contribution is the systematic evaluation across cell lines and degradation types, revealing the quality-dependence of filter performance. Previous microscopy studies compared at most 2–3 filter types on a single cell line; our 12-filter, 8-cell-line evaluation is the most comprehensive to date.

Practical recommendations. Based on our results, we recommend: (1) DoG ($\sigma_1=0.05$, $\sigma_2=0.20$) as the default for segmentation preprocessing; (2) Homomorphic ($\gamma_L=0.3$, $\gamma_H=2.5$) for images with illumination gradients; (3) Butterworth ($n=2$) as the safest general-purpose choice; (4) avoiding Ideal and Elliptic filters for biological images (ringing around cell boundaries); (5) implementing cell-line-adaptive filter selection for production systems.

Limitations. (1) Evaluation is limited to phase-contrast microscopy; results may not transfer to fluorescence or DIC. (2) Otsu thresholding, while standard, is a simple segmentation method; more sophisticated methods (Cellpose, StarConvex) may show different filter preferences. (3) The 25% annotation subset limits some analyses. (4) Synthetic degradations may not capture all real-world quality issues.

6 Conclusion

We presented the first systematic evaluation of 12 bandpass filter types for phase-contrast microscopy segmentation across 8 cell lines and 13 degradation types, totaling over 20,000 segmentation evaluations. Key contributions:

1. No single filter is universally optimal—the best filter depends on cell line and image quality
2. Filter improvements on LQ images are 10–100× smaller than on HQ images
3. Filter transfer from HQ to LQ is consistently below 15%
4. Adaptive, cell-line-specific selection improves mean IoU by +0.130

5. Evidence-based, degradation-specific filter recommendations

All code, trained models, and the adaptive selection algorithm are available at https://github.com/mjonyh/microscopic_images.

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